EECS 16A Designing Information Devices and Systems I
Homework 3

Submission Format
Your homework submission should consist of **one** file.

- `hw3.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit each file to its respective assignment on Gradescope.

1. Reading Assignment

For this homework, please read Notes 4 through 6. Note 4 will discuss mathematical derivation techniques useful for proofs. The notes 5 and 6 will provide an overview of multiplication of matrices with vectors, by considering the example of water reservoirs and water pumps, and matrix inversion.

(a) Please summarize (in one or two paragraphs) the main learnings in Notes 4-6.

(b) You have seen in Note 5 that the pump system can be represented by a state transition matrix. What constraint must this matrix satisfy in order for the pump system to obey water conservation?

(c) Repeat the proof of Theorem 6.4 in Note 6. That is, prove that **If a matrix A is invertible, its columns are linearly independent.**

2. Study Groups Tasks

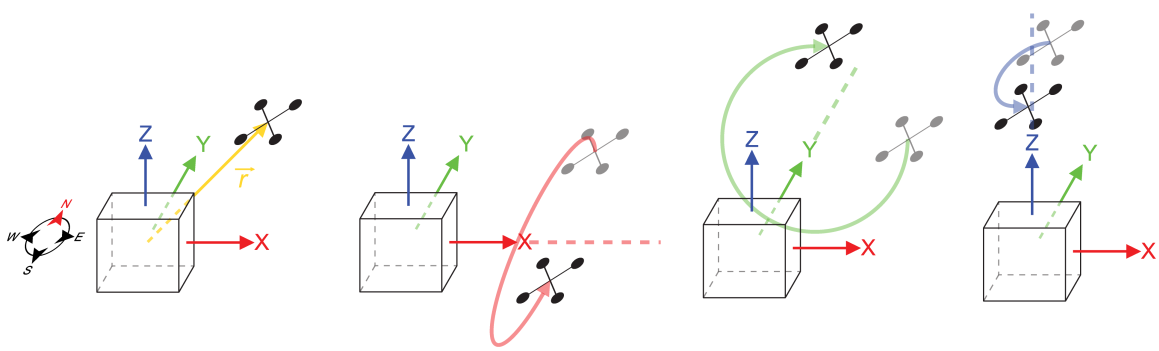
(a) We recommend you use some time during your study group to develop a plan for studying for the first midterm, which is coming soon. Discuss how you can support each other in your studying. Do you want to have an extra meeting to discuss some harder concepts? If you are not working with a group, you may make your plan alone. Make a list of what you want to review — this should include lectures, discussion problems, homework problems, notes.

(b) We hope your study groups have been productive and fun so far! In case you missed the first study group matching form and have not connected with a group yet, or you would like to try a new study group, we are conducting a second matching process. Please use the link below to make your request.
`https://forms.gle/Man8tZskVJyne7Pu8`
To get full credit for this question, you must (1) fill out the survey (it will record your email) and (2) indicate in your HW submission that you filled out the survey.

3. Quadcopter Transformations

Learning Objectives: *Linear algebra is often used to represent transformations in robotics. This problem introduces some of the basic uses of transformations.*

Sunash and his colleagues are interested in developing a communication link using a laser to control the location of a quadrotor. Consider a vector $\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3$ representing the location of the quadcopter relative to the origin. The quadcopter is only capable of three different maneuvers relative to the origin. The maneuvers are rotations about the x, y, and z axes. For perspective, the positive x-axis points east, the positive y-axis points north, and the positive z-axis points up towards the sky. The figures below illustrate the quadcopter and these maneuvers.



We can represent each of these rotations, that are linear transformations, as matrices that operate on the location vector of the quadcopter, $\vec{r}$, to position it at its new location. The matrices $\mathbf{R_x}(\theta)$, $\mathbf{R_y}(\psi)$, and $\mathbf{R_z}(\phi)$ represent rotations about the x-axis, y-axis, and z-axis, respectively. The matrices are:

$$\mathbf{R_x}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \mathbf{R_y}(\psi) = \begin{bmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{bmatrix}, \mathbf{R_z}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

.

(a) Sunash wants to make the quadcopter rotate first by 30° about the x-axis, and then by 60° about the z-axis. Use $\mathbf{R_x}(\theta)$, $\mathbf{R_y}(\psi)$, and $\mathbf{R_z}(\phi)$ to construct **a single matrix that performs the operations in the specified order**. Show the matrix operations and calculations by hand.

(b) Sunash accidentally punched in the two rotation commands in reverse. The 60° rotation about the z-axis occurred before the 30° rotation about the x-axis. Use $\mathbf{R_x}(\theta)$, $\mathbf{R_y}(\psi)$, and $\mathbf{R_z}(\phi)$ to construct **a single matrix that performs the operations in the accidentally reversed order**. Show the matrix operations and calculations by hand.

(c) Say the quadcopter was initially positioned at $\vec{r} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$.

   i. Where did Sunash intend for the quadcopter to end up? Use your result from part (a) to find this out.

   ii. Where did the quadcopter actually end up with the accidentally reversed order of the rotations? Use your result from part (b) to find this out.

   iii. Did the quadcopter end up where it was supposed to go?

4. Mechanical Inverses

Learning Objectives: *Matrices represent linear transformations, and their inverses represent the opposite transformation. Here we practice inversion, but are also looking to develop an intuition. Visualizing the transformations might help develop this intuition.*

For each of the following values of matrix A:

i Find the inverse, $\mathbf{A}^{-1}$, if it exists. If you found that the inverse does not exist, mention how you decided that. Solve this by hand.

ii **For parts (a)-(d)**, in addition to finding the inverse (if it exists), describe how the matrix, $\mathbf{A}$ transforms an arbitrary vector, $\begin{bmatrix} x \\ y \end{bmatrix}$.

For example, if $\mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$, then $\mathbf{A}$ could scale $\begin{bmatrix} x \\ y \end{bmatrix}$ by 2 to get $\begin{bmatrix} 2x \\ 2y \end{bmatrix}$. If $\mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$, then $\mathbf{A}$ could reflect $\begin{bmatrix} x \\ y \end{bmatrix}$ across the x axis, etc. *Hint:* It may help to plot a few examples to recognize the pattern.

iii **For parts (a)-(d)**, if we use $\mathbf{A}$ to geometrically transform $\begin{bmatrix} x \\ y \end{bmatrix}$ to get $\begin{bmatrix} u \\ v \end{bmatrix} = \mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix}$, **is it possible to reverse the transformation geometrically,** i.e. is it possible to retrieve $\begin{bmatrix} x \\ y \end{bmatrix}$ from $\begin{bmatrix} u \\ v \end{bmatrix}$ geometrically?

(a) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) **(PRACTICE / NOT GRADED)**
$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

(d) **(PRACTICE / NOT GRADED)**
$\mathbf{A} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$
Assume $\cos\theta \neq 0$. *Hint:* $\cos^2\theta + \sin^2\theta = 1$.

(e) $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$

(f) $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 1 & 4 & 4 \end{bmatrix}$

(g) **(PRACTICE / NOT GRADED)**
$\mathbf{A} = \begin{bmatrix} -1 & 1 & -\frac{1}{2} \\ 1 & 1 & -\frac{1}{2} \\ 0 & 1 & 1 \end{bmatrix}$

(h) **(PRACTICE / NOT GRADED)**
$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

(i) **(PRACTICE / NOT GRADED)**

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & -2 & 1 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 0 & 4 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Hint 1: What do the linear (in)dependence of the rows and columns tell us about the invertibility of a matrix? Hint 2: We're reasonable people!

5. Properties of Pump Systems

Learning Objectives: *This problem illustrates how matrices and vectors can be used to represent linear transformations.*

Throughout this problem, we will consider a system of reservoirs connected to each other through pumps. An example system is shown below in Figure 1, represented as a graph. Each node in the graph is marked with a letter and **represents a reservoir**. Each arrow in the graph represents a pump which moves a fraction of the water from one reservoir to the next at every time step. The **fraction of water** is written on top of the arrow.

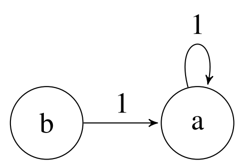


Figure 1: Pump system

(a) Consider the system of pumps shown above in Figure 1. Let $x_a[n]$ and $x_b[n]$ represent the amount of water in reservoir a and b, respectively, at time step n. Find a system of equations that represents $x_a[n+1]$ and $x_b[n+1]$ in terms of $x_a[n]$, $x_b[n]$ etc.

(b) For the system shown in Figure 1, find the associated state transition matrix. In other words, find the matrix $\mathbf{A}$ such that:

$$\vec{x}[n+1] = \mathbf{A}\vec{x}[n], \text{ where } \vec{x}[n] = \begin{bmatrix} x_a[n] \\ x_b[n] \end{bmatrix}$$

(c) Let us assume that at time step 0, the reservoirs are initialized to the following water levels: $x_a[0] = 0.5, x_b[0] = 0.5$. In a completely alternate universe, the reservoirs are initialized to the following water levels: $x_a[0] = 0.3, x_b[0] = 0.7$. For both initial states, what are the water levels at timestep 1 ($\vec{x}[1]$)? Use your answer from part (b) to compute your solution.

(d) If you observe the reservoirs at timestep 1, i.e if you know $\vec{x}[1]$, can you figure out what the initial ($\vec{x}[0]$) water levels were? Why or why not?

(e) Now let us generalize what we observed. Say there is a transition matrix $\mathbf{A}$ representing a pump system. Say there exist two distinct initial state vectors/ water levels: $\vec{x}_u[0]$ and $\vec{x}_v[0]$, that lead to the same state vector $\vec{x}[1]$ after $\mathbf{A}$ acts on them.You do not know which of the two initial state vectors you started in. Can you decide which initial state you started in by observing $\vec{x}[1]$? What does this say about the matrix $\mathbf{A}$?

(f) Now, we want to prove the following theorem in a step-by-step fashion.

Theorem: Consider a system consisting of k reservoirs such that the entries of each column in the system's state transition matrix sum to one. If s is the total amount of water in the system at timestep n, then total amount of water at timestep $n+1$ will also be s.

i. Rewrite the theorem statement for a graph with only two reservoirs.

ii. Since the problem does not specify the transition matrix, let us consider the transition matrix $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and the state vector $\vec{x}[n] = \begin{bmatrix} x_1[n] \\ x_2[n] \end{bmatrix}$. In general, it is helpful to write as much out mathematically as you can in proofs. It can also be helpful to draw the transition graph. Write out what is "known", i.e. ALL the information that is given to you in the theorem statement *in mathematical form*.

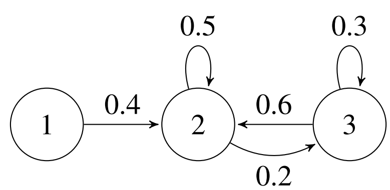
iii. Now write out what is to be proved *in mathematical form*.

iv. Prove the statement i.e. the ans for your last part for the case of two reservoirs.

v. Now use what you learned to generalize to the case of k reservoirs. *Hint:* Think about **A** in terms of its columns, since you have information about sum of each column.

(g) Set up the state transition matrix **A** for the system of pumps shown below. Compute the sum of the entries of each column of the state transition matrix. Are the sums greater than/less than/equal to 1? Explain what this **A** matrix physically implies about how the total amount of water in this system changes over time.

Note: Arrows not shown assumed to be 0.



6. Segway Tours

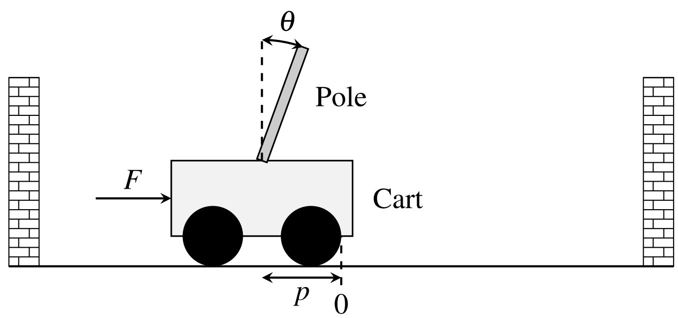
Learning Objective: *The learning objective of this problem is to see how the concept of span can be applied to control problems. If a desired state vector of a linear control problem is in a span of a particular set of vectors, then the system may be steered to reach that particular vector using the available inputs.*

Your friend has decided to start a new SF tour business, and you suggest they use segways.

They become intrigued by your idea and asks you how a segway works. A segway is essentially a stand on two wheels.

The segway works by applying a force (through the spinning wheels) to the base of the segway. This controls both the position on the segway and the angle of the stand. As the driver pushes on the stand, the segway tries to bring itself back to the upright position, and it can only do this by moving the base.

Is it possible for the segway to be brought upright and to a stop from any initial configuration? There is only one input (force) used to control two outputs (position and angle). You both talk to a third friend who is GSIing EE128, and she tells you that a segway can be modeled as a cart-pole system.



A cart-pole system can be fully described by its position p, velocity $\dot{p}$, angle θ, and angular velocity $\dot{\theta}$. We write this as a "state vector", $\vec{x}$:

$$\vec{x} = \begin{bmatrix} p \\ \dot{p} \\ \theta \\ \dot{\theta} \end{bmatrix}.$$

The input to this system is a scalar quantity $u[n]$ at time n, which is the force F applied to the cart (or base of the segway).[1]

The cart-pole system can be represented by a linear model:

$$\vec{x}[n+1] = \mathbf{A}\vec{x}[n] + \vec{b}u[n], \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{4\times 4}$ and $\vec{b} \in \mathbb{R}^{4\times 1}$.

The control $u[n]$ allows us to move the state ($\vec{x}$) in the direction of $\vec{b}$. So, if $u[n] = 2$, we move the state by $2\vec{b}$ at time n, and so on. We can choose different controls at different times.

The model tells us how the the state vector, $\vec{x}$, will evolve over time as a function of the current state vector and control inputs.

You look at this general linear system and try to answer the following question: Starting from some initial state $\vec{x}_0$, can we reach a final desired state, $\vec{x}_f$, in N steps?

The challenge seems to be that the state is four-dimensional and keeps evolving and that we can only apply a one-dimensional (scalar) control at each time. Typically, to set the values of four variables to desired quantities, you would need four inputs. Can you do this with just one input?

We will solve this problem by walking through several steps.

(a) Express $\vec{x}[1]$ in terms of $\vec{x}[0]$ and the input $u[0]$. (*Hint:* This is not meant to be complicated.)

(b) i. Express $\vec{x}[2]$ in terms of *only* $\vec{x}[0]$ and the inputs, $u[0]$ and $u[1]$.

ii. Then express $\vec{x}[3]$ in terms of *only* $\vec{x}[0]$ and the inputs, $u[0]$, $u[1]$, and $u[2]$.

[1] You might note that velocity and angular velocity are derivatives of position and angle respectively. Differential equations are used to describe continuous time systems, which you will learn more about in EECS 16B. But even without these techniques, we can still approximate the solution to be a continuous time system by modeling it as a discrete time system where we take very small steps in time. We think about applying a force constantly for a given finite duration and we see how the system responds after that finite duration.

iii. Finally express $\vec{x}[4]$ in terms of *only* $\vec{x}[0]$ and the inputs, $u[0]$, $u[1]$, $u[2]$, and $u[3]$. **Your expressions can have other relevant variables (e.g. A, $\vec{b}$ etc) and mathematical operators.**

(c) Now, generalize the pattern you saw in the earlier part to write an expression for $\vec{x}[N]$ in terms of $\vec{x}[0]$ and the inputs from $u[0],\ldots,u[N-1]$. **Your expression can have other relevant variables (e.g. A, $\vec{b}$ etc) and mathematical operators.**

For the next four parts of the problem, you are given the matrix **A** and the vector $\vec{b}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 0.05 & -0.01 & 0 \\ 0 & 0.22 & -0.17 & -0.01 \\ 0 & 0.10 & 1.14 & 0.10 \\ 0 & 1.66 & 2.85 & 1.14 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 0.01 \\ 0.21 \\ -0.03 \\ -0.44 \end{bmatrix}$$

Assume the cart-pole starts in an initial state $\vec{x}[0] = \begin{bmatrix} -0.3853493 \\ 6.1032227 \\ 0.8120005 \\ -14 \end{bmatrix}$, and you want to reach the desired state $\vec{x}_f = \vec{0}$ using the control inputs $u[0], u[1], \ldots$ etc. The state vector $\vec{x}_f = \vec{0}$ corresponds to the cart-pole (or segway) being upright and stopped at the origin. **Reaching $\vec{x}_f = \vec{0}$ in N steps means that, given that we start at $\vec{x}[0]$, we can find control inputs ($u[0], u[1], \ldots$ etc), such that we get $\vec{x}[N]$ (i.e. state vector at Nth time step) equal to $\vec{x}_f = \vec{0}$.**

Note: Please use the Jupyter notebook to solve parts (d) - (g) of the problem. You may use the function we provided `gauss_elim(matrix)` to help you find the **upper triangular form** of matrices. An example of Gaussian Elimination using (`gauss_elim(matrix)`) is provide in the Jupyter notebook under section **Example Usage of gauss_elim**. You may also use the function (`np.linalg.solve`) to solve the equations.

(d) Can you reach $\vec{x}_f$ in *two* time steps? Show work to justify your answer. You should manipulate the equations on paper, but then use the Jupyter notebook for numerical computations.
(*Hint: Express $\vec{x}[2] - \mathbf{A}^2\vec{x}[0]$ in terms of the inputs $u[0]$ and $u[1]$. Then determine if the system of equations can be solved to obtain $u[0]$ and $u[1]$. If we obtain valid solutions for $u[0]$ and $u[1]$, then we can say we will reach $\vec{x}_f$ in two time steps. Use the notebook to see if the system of equations can be solved.*)

(e) Can you reach $\vec{x}_f$ in *three* time steps? Show work to justify your answer. You should manipulate the equations on paper, but then use the Jupyter notebook for numerical computations.
(*Hint: Similar to the last part, express $\vec{x}[3] - \mathbf{A}^3\vec{x}[0]$ in terms of the inputs $u[0]$, $u[1]$ and $u[2]$. Then determine if we can obtain valid solutions for $u[0]$, $u[1]$ and $u[2]$.*)

(f) Can you reach $\vec{x}_f$ in *four* time steps? Show work to justify your answer. You should manipulate the equations on paper, but then use the Jupyter notebook for numerical computations. (*Use the hints from the last two parts.*)

(g) If you have found that you can get to the final state in 4 time steps, find the required correct control inputs, i.e. $u[0]$, $u[1]$, $u[2]$ and $u[3]$, using Jupyter and verify the answer by entering these control inputs into the **Plug in your controller** section of the code in the Jupyter notebook. You need to just show

that you reached the desired final state $\vec{x}_f$ by plugging in the control inputs. The code has been already written to simulate this system.
Suggestion: See what happens if you enter all four control inputs equal to 0. This gives you an idea of how the system naturally evolves!

(h) Let us reflect on what we just did. Recall the system we have:

$$\vec{x}[n+1] = \mathbf{A}\vec{x}[n] + \vec{b}u[n].$$

The control allows us to move the state at time step $n+1$ by $u[n]$ in direction $\vec{b}$, remember $u[n] \in \mathbb{R}$ is just a scalar. We know from part (c) that:

$$\vec{x}[2] = \mathbf{A}^2\vec{x}[0] + \mathbf{A}\vec{b}u[0] + \vec{b}u[1].$$

Again, here $u[0], u[1] \in \mathbb{R}$ can be thought of as arbitrary scalars, and $\mathbf{A}\vec{b}u[0] + \vec{b}u[1]$ can be thought of as the set of all linear combinations of the vectors $\vec{b}$ and $\mathbf{A}\vec{b}$. Using this observation, can you express the possible states you can arrive at in two time steps using the span of exactly *two* vectors plus a vector offset?

(i) Let's try to generalize the idea in the previous part. Express the states you can reach in N timesteps as a span of some vectors plus a vector offset. (Hint: Consider the direction that each control input $u[0], \ldots, u[N-1]$ can move $\vec{x}[N]$ by.)

(j) **(CHALLENGE, OPTIONAL)** Now say you wanted to reach anywhere in $\mathbb{R}^4$, i.e. $\vec{x}_f$ is an unspecified vector in $\mathbb{R}^4$. Under what conditions can you guarantee that you can "reach" $\vec{x}_f$ from any $\vec{x}_0$ in N time steps?
Wouldn't this be cool?

7. Discussion Opt-In/Opt-Out Form

Here is the form for indicating whether you'd like to opt-in or opt-out of discussion participation. For those who've already filled it out, great! For those who haven't, the deadline is now extended to September 22nd at 11:00PM PT. If you've already filled out the form but would like to change your decision, you may edit your response before the deadline. After the deadline passes, your decision is binding, so please read the section "Participation in Discussion" carefully before deciding. If you do not fill it out, we will assume that you are opting-out.

8. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.